Sr.No. 8501

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:401 Course Section:1	Seat No: 2022600031

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Name of the candidate. MAAZ MALIK

Candidate's UID number: 2 0 2 2 6 0 0 0 3 1

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE AIML

DAY AND DATE: 08/10/25 Wed

COURSE NAME: DEEP LEARNING

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

1028

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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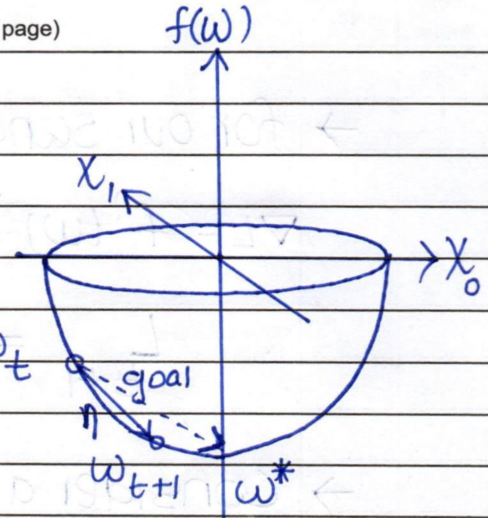

[QUESTION 1]
[A]

$$f(\omega_t) = \frac{1}{2} \| X\omega_t - y \|^2$$

↑ Loss function L

Consider we are currently at ω_t and want to update position to ω_{t+1}, we will move against the direction of the gradient towards ω^* (optimal point). lets say we move a small step $\Delta\omega$.

General scenario

$$f(\omega + \Delta\omega) = f(\omega) + f'(\omega)\Delta\omega + \frac{f''(\omega)\Delta\omega^2}{2!} + \frac{f'''(\omega)\Delta\omega^3}{3!} + \dots \text{so on}$$

... using taylor series

lets consider first two terms as the other terms contribute very small values.

$$f(\omega + \Delta\omega) = f(\omega) + f'(\omega)\Delta\omega + 0$$

We know $f(\omega)$ indicates loss

$$\underset{\text{new loss}}{L_{t+1}} = \underset{\text{old loss}}{L_t} + \underset{\text{gradient or slope}}{\nabla L} \cdot \Delta L \leftarrow \text{Small step } (\eta)$$

our goal during gradient descent is to minimize loss ie. L ie $f(\omega)$. t denotes the training step/iteration.

$$L_{t+1} = L_t - \nabla L \, \eta \leftarrow \text{also known as learning rate}$$

↑ moving against the gradient

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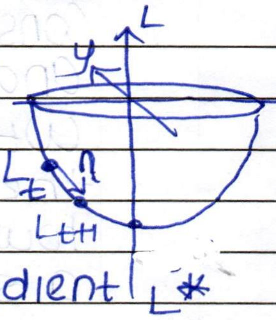
→ for our scenario

$$\nabla L = f'(w) = \frac{1}{2} \times 2 \times x = x \quad \text{...derivating wrt } w$$

$$L_{t+1} = L_t - \overbrace{\eta \underbrace{x}_{\text{gradient}}}^{\text{step}}$$

→ Consider a general quadratic loss curve in 3 Dimensions

$$L = f(x, y) = \frac{1}{2}(ax^2 + by^2)$$



$$\nabla L = f'(x, y) = \begin{bmatrix} ax \\ by \end{bmatrix} \quad \text{...gradient}$$

$$\nabla^2 L = f''(x, y) = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \quad \text{...heissian}$$

→ a and b are also eigen values for this case

$$x_{t+1} = x_t - \eta a x_t \quad \text{...from gradient update eq}^n$$

$$y_{t+1} = y_t - \eta b y_t$$

$$\Downarrow$$

$$x_{t+1} = x_t(1 - \eta a)$$

$$y_{t+1} = y_t(1 - \eta b)$$

for $x_{t+1} < x_t$ ...ie. Convergence → we need to reduce x_t on every step t

$0 < (1 - \eta a) < 1$
$0 < (1 - \eta b) < 1$ → $0 < \eta < \frac{2}{\underbrace{\max(a, b)}_{\lambda max}}$

→ think of a and b as eccentricity of curve along x and y axis

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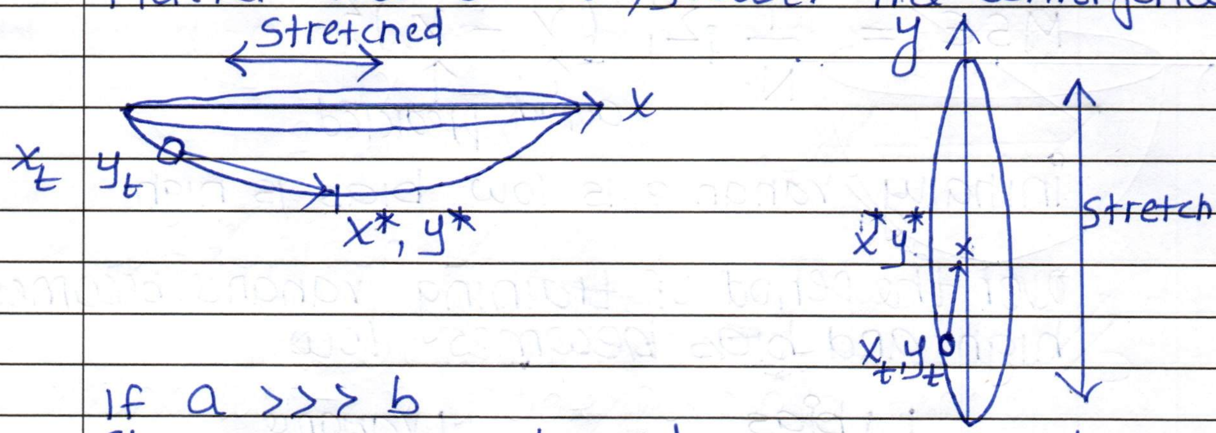
→ Steeper the curve, faster the convergence
flatter the curve, slower the convergence.



If $a >>> b$
Slow convergence based on a

If $a <<< b$
Slow convergence based on b

∴ We take whatever is maximum as it determines rate of convergence
ie max(a, b)

$a, b \leftrightarrow \lambda_{min}, \lambda_{max}$

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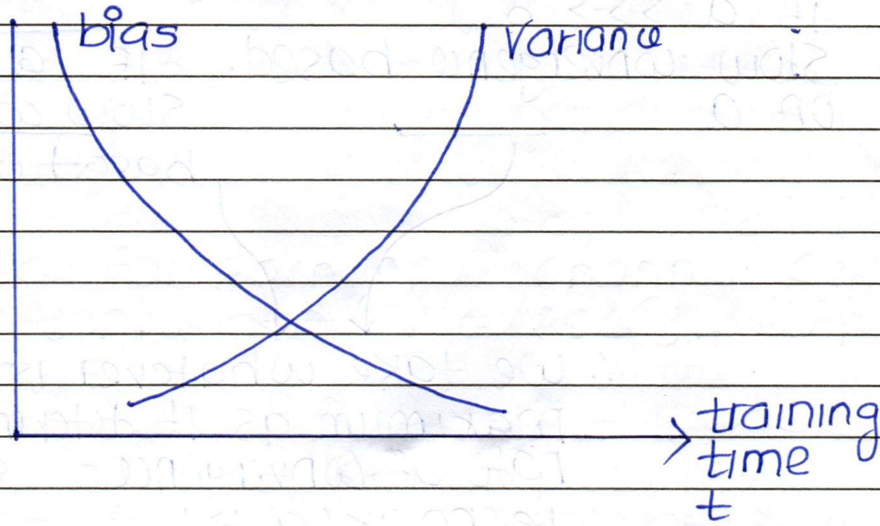
[QUESTION 1B]

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (\underset{\text{actual}}{\underset{\uparrow}{x}} - \underset{\text{predicted}}{\underset{\uparrow}{\hat{x}}})^2$$

initially variance is low bias is high

Over the period of training variance becomes high and bias becomes low



At some point during training it happens so that variance and bias are optimal before which underfitting happens and above which overfitting happens.

when model complexity is high, the model captures noise as a feature. This leads to overfitting, as model starts getting trained on noise instead of the actual features.

This leads to poor generalization.

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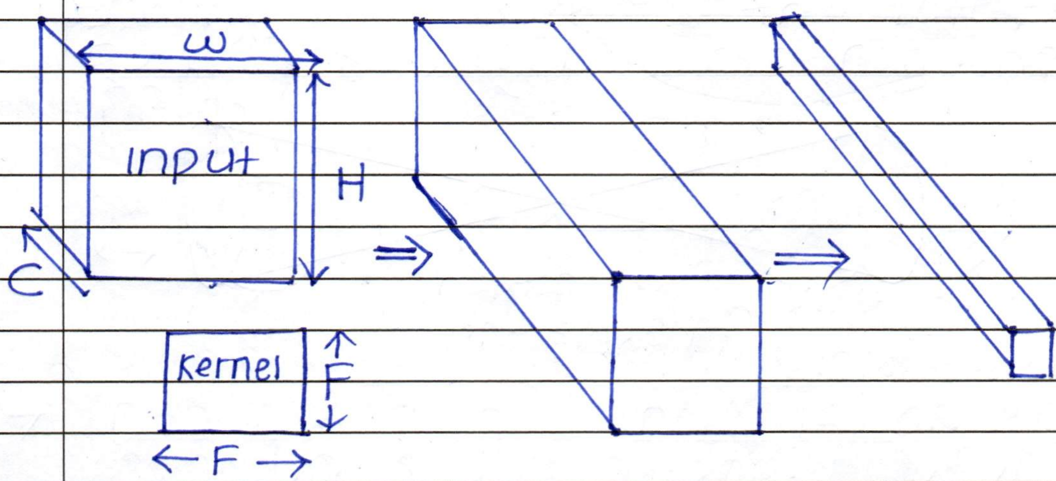
[QUESTION 2 A]

Consider a normal convolution network



lets say for each input channel C_{in}, Kernel convolution produces C_{out} output channels.

for each input, Kernal performs K x K dot products.

$$\text{Size of output image} = \left(\left(\frac{\overset{\text{input size}}{N} + \overset{\text{Padding}}{2P} - \underset{\text{kernel size}}{K}}{2}\right) + 1\right)^2$$

Total no. of dot product operations in a single layer =

$$\underbrace{\left(\frac{W+2P-F}{2}+1\right)}_{\text{for width}}\underbrace{\left(\frac{H+2P-F}{2}+1\right)}_{\text{for height}} \times \overset{\text{for each channel inputted}}{C_{in}} \times \underset{\text{kernel area}}{F^2} \times \underset{\text{for each input channel } C_{out} \text{ output channels}}{C_{out}}$$

for L layers

~~log~~

$$\cancel{\left(W \; C^{out} \left(\frac{W+2P-F}{2}+1\right)\left(\frac{H+2P-F}{2}+1\right)\left(C_{in} C_{out} F^2\right)\right)}$$

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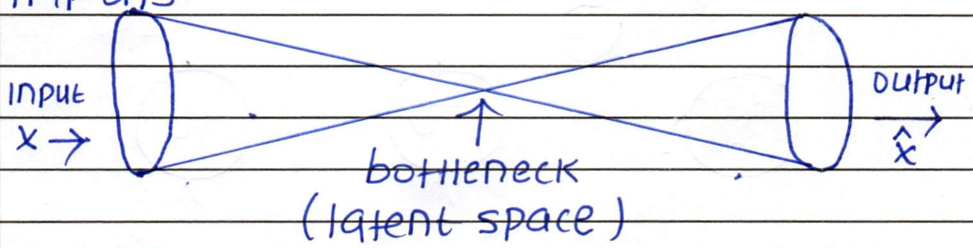
[QUESTION 2B]

→ undercomplete auto-encoder means that the no. of neurons in the latent layer/bottleneck layer is less than the no. of inputs



→ This implies that there is an effective reduction in dimensionality of vectors causing the representation of input to occur in a lower latent space.

The net effect is dimentionality reduction - The same thing PCA does.

If input size was 10 and hidden layer size is 2, ~~A~~ 10 dimensional vector is compressed/flattened/reduced down to a two dimensional vector in a two dimensional latent space.

PCA by nature is linear in nature. It just performs rotation of the coordinates to find a view/projection that best seprates the data points.

when trained with MSE, It simply learns the linear difference

$$(x - \hat{x})$$

between the input image and the output image. Hence it will behave equivalent to PCA.

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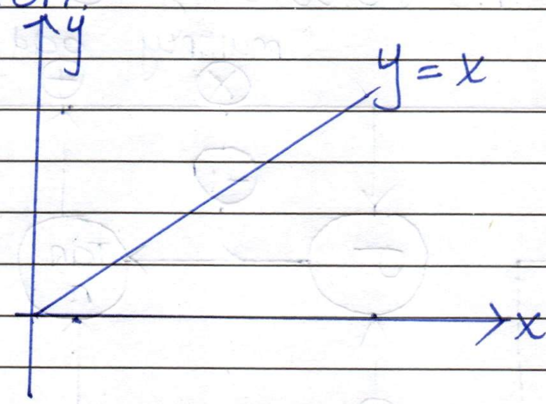
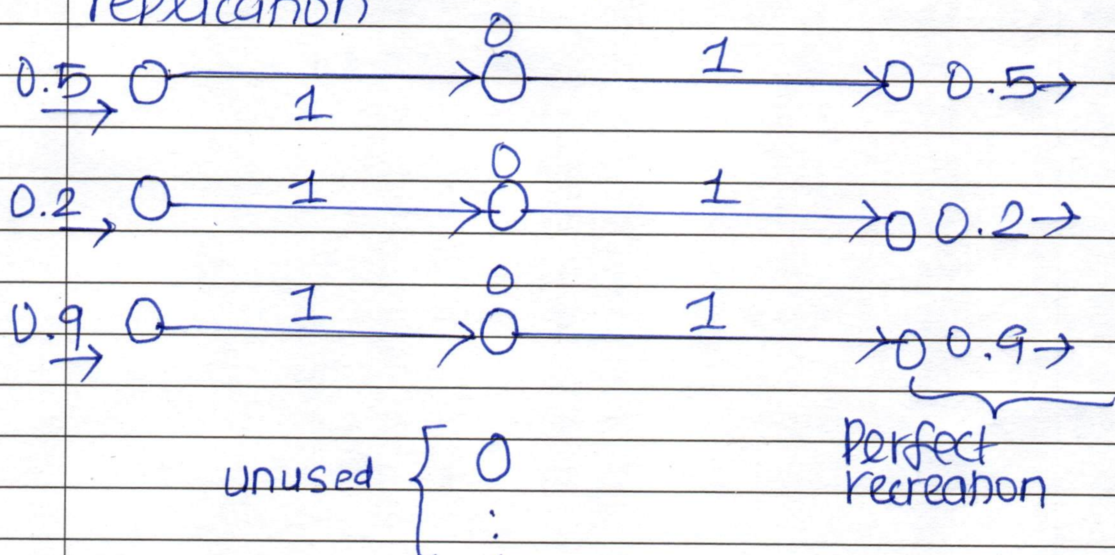
→ An overcomplete autoencoder without regularization tends to learn the identity function.



this is possible because one to one mapping is possible between neurons in input, and latent and output.

Each neuron makes it a responsibility to pass out the same value as inputted as this will also produce the same input replication



→ all weights become one and any cross links will die out.

→ Regularization solves this problem by shutting off most neurons in the latent or bottleneck layer.

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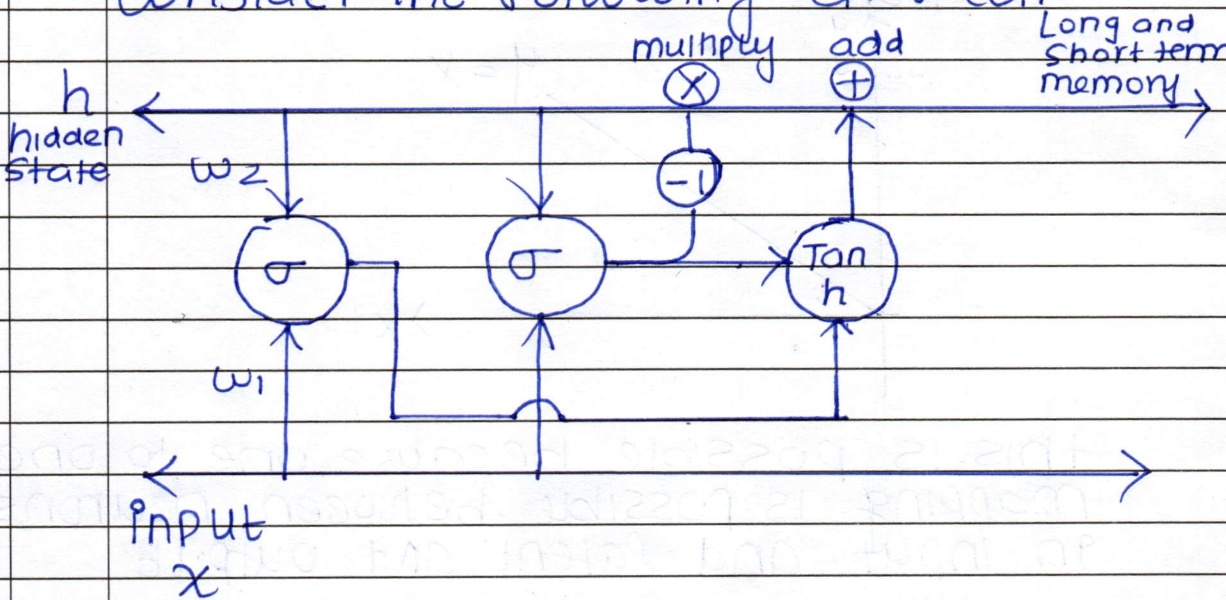
[QUESTION 3B]

Consider the following GRU cell



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